
Discussion:

Pressured to Choose? Firms' product choices under the EU ETS

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Research Question

- How does the EU ETS affect firms' product choices?

Method

- Link German **firm-product data** to EU ETS treatment status
- Estimate a **matched difference-in-differences** design
- Develop a **multi-product firm model** to understand mechanisms

Findings and Contribution

1. EU ETS increases product dropping, especially energy-intensive products
2. Product mix contributes only modestly to emission-intensity reductions
3. Multi-product model highlights product choice as an adjustment margin

Identification

Problem

- Treatment is based on EU ETS coverage, but assignment is not random
- **Ideally:** comparing firms with installations just around the 20 MW cutoff
- **Alternative matching variable:** maximum plant-level energy use

Why it is useful?

Firm	ETS treated	Max plant energy use	Interpretation
A	Yes	100 GWh/year	Treated firm
B	No	100 GWh/year	Good control
C	No	5 GWh/year	Bad control

⇒ **A and B better to compare than A and C**

Remaining Concern: Utilization

Firm	ETS treated	Capacity	Utilization	Max plant energy use
A	Yes	High	Medium	100 GWh/year
B	No	Medium	Very high	100 GWh/year

- Treatment is assigned based on 20MW **capacity** cutoff
- Same energy use with high capacity (Treatment): **High utilization**
- Low utilization may be correlated with outcomes, e.g.
 1. Weaker demand: more likely to drop products
 2. Excess capacity: predictor for restructuring
 3. Older: less efficient, or less competitive
- **Suggestion:** Add for largest plant $\frac{\text{Average Energy Use}}{\text{Max Energy Use Observed}}$

Remaining concern: Process Technology

Firm	ETS	Energy use	Process
A	Yes	100 GWh	Upstream processing
B	No	100 GWh	Downstream processing

- More likely to cross 20 MW threshold conditional on energy:
 - Upstream
 - Combustion-based
 - Kiln-based or furnace-based
- Upstream firms may have:
 - Fewer, larger and standardized product lines: less flexible ($\downarrow$ bias)
 - Maybe also restructuring pressure, older capital, declining demand ($\uparrow$ bias)
- **Suggestion:**
 1. Add NACE 3- or NACE 4-by-year fixed effects
 2. Conditional on similar baseline products

Remaining concern: Fuel mix

Firm	ETS treated	Energy use	Fuel mix	Direct exposure
A	Yes	100 GWh	Fossil fuels	High
B	No	100 GWh	Electricity	Lower

- **On-site fossil combustion** relevant for 20 MW threshold
- Firms using mostly electricity often not regulated conditional on energy use
- Treatment may correlate with fossil-combustion firms, e.g. which could be
 1. More upstream
 2. More capital-intensive
 3. More exposed to restructuring
- **Suggestion:** show balance and robustness on fuel mix